

KENOSHA LH, WI, USA

WMO: 997340

Lat: 42.5888N

Lon: 87.8085W

Elev: 643

StdP: 14.36

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.2	(c) 1.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 30.7	(k) 26.5	(l) 28.3	(m) 25.5	(n) 13.8	(o) 280	(p) 0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.3	(c) 87.5	(d) N/A	(e) 83.7	(f) N/A	(g) 80.2	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 13.9	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 30.9	(b) 26.7	(c) 24.0	(e) -9.7	(f) 92.8	(g) 6.5	(h) 4.2	(i) -14.3	(j) 95.8	(k) -18.1	(l) 98.3	(m) -21.8	(n) 100.7	(o) -26.5	(p) 103.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	22.7	23.8	34.6	44.5	53.9	64.3	71.4	70.8	63.8	52.6	40.2	30.0
	DBStd	18.93	12.19	10.84	9.49	6.76	7.37	7.01	6.13	4.85	6.64	7.74	9.95	10.12
	HDD50	3302	846	734	486	195	41	0	0	0	1	63	317	619
	HDD65	6807	1311	1154	941	615	352	97	14	10	98	387	745	1083
	CDD50	2524	0	0	10	29	163	429	664	645	416	145	22	1
	CDD65	556	0	0	1	0	9	76	213	190	63	4	0	0
	CDH74	2790	0	0	1	7	100	407	1257	745	256	17	0	0
	CDH80	740	0	0	0	1	24	99	392	168	55	1	0	0
(14) Wind	WSAvg	11.7	12.8	12.9	11.8	13.5	11.7	10.0	9.5	9.6	11.0	12.2	12.8	12.6
(15) Precipitation	PrecAvg	34.6	1.7	1.5	2.2	3.7	3.8	4.1	3.3	3.7	3.2	2.8	2.6	1.9
	PrecMax	41.6	3.7	3.5	4.4	7.1	10.6	8.5	7.3	11.2	6.9	6.5	5.4	4.1
	PrecMin	24.0	0.4	0.2	0.6	1.8	0.6	0.7	1.7	1.4	0.7	0.9	0.6	0.6
	PrecStd	4.6	0.9	0.9	1.2	1.5	2.1	2.0	1.3	1.9	1.5	1.5	1.4	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	53.5	69.4	72.8	83.5	89.7	95.1	89.7	86.0	76.5	65.6	57.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	47.9	45.6	59.9	65.6	76.7	83.6	88.5	85.0	80.3	70.7	61.0	49.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	42.4	40.7	52.4	59.1	71.3	79.4	84.5	80.6	76.5	66.3	56.9	45.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	38.8	36.7	46.0	53.6	65.9	75.5	80.7	77.7	73.2	63.4	53.1	42.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	13.1	12.5	10.9	11.1	13.1	12.3	11.3	10.3	11.5	11.9	11.7	11.5
		MCDBR	17.0	16.8	20.3	21.7	23.6	18.7	17.7	15.1	16.1	16.3	16.2	13.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.293	0.310	0.343	0.394	0.419	0.437	0.435	0.428	0.399	0.351	0.313	0.292
	taud		2.450	2.401	2.397	2.287	2.268	2.267	2.272	2.291	2.345	2.476	2.524	2.491
	Ebn at Noon		271	283	286	277	272	266	266	264	263	265	262	261
	Edh at Noon		24	29	33	40	42	42	42	40	35	27	22	21
(44) All-Sky Solar Radiation	RadAvg		462	742	1123	1456	1675	1917	1947	1682	1364	881	548	382
	RadStd		61	88	95	121	131	127	88	104	116	100	55	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page